Finite Menus of Spectral Subspaces: Realizability and the Cost of Convexification

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Abstract

We study fixed menus of rank-r orthogonal subspaces under a shared quadratic reward and multiple quadratic resource costs, with worst-case additive regret over a compact scalar-price interval. The fully adaptive value curve determines a convexified menu problem but can conceal different realizable optima. We characterize exactly when this relaxation is valid on every price subinterval: the slopes of actual optimal presets must fill the superdifferential interval at every interior price. For spectral menus, this becomes a union-of-intervals test on active cutoff eigenspaces and always holds for a single site. If q prices fail the criterion, every relaxed K-menu admits a replacement by at most $K + \min(K, q)$ globally price-optimal presets at no greater regret. Analytic eigenvalue parameterization makes q finite for each fixed spectral instance. The resulting additive cardinality bound transfers any separately established positive power-law asymptotic. Sharp spectral examples also realize the factor-two comparison supplied by endpoint scalarization. We combine established two-criterion spectral optimization and interval covering with reusable bounds and solver-independent outward verification. Synthetic dense and sparse experiments demonstrate construction and continuous-price certification, including banded instances of dimension 4096, while documenting regimes where specialized or dense methods are faster. The contribution is the realizability criterion, its spectral characterization, and the exceptional-price refinement of finite-menu repair.

Keywords: minimax regret; spectral optimization; finite adaptability; subspace menus; interval verification.

1 Introduction

A continuously adjustable measurement subspace can be optimal for each resource price, but a device or procedure may instead offer only a fixed number of presets. Choosing those presets is a different optimization problem. A good menu must control its worst loss relative to full adaptation, including prices between any sampled design points. We consider a quadratic model in which a preset is an orthogonal subspace, the task reward is shared across sites, and the resource cost depends on both the site and the subspace. The price is a single nonnegative scalar. These assumptions give an especially useful interval geometry without making the collection of realizable preset lines convex across sites.

This paper studies what is lost by replacing realizable presets with scalar approximating lines. A family with an unchanged adaptive curve has a continuum of different one-preset losses. More generally, missing slopes on globally optimal faces exactly characterize failure of the scalar relaxation on some subinterval. Those exceptional prices also bound the extra number of globally price-optimal

presets needed to match a relaxed menu. This makes the distinction quantitative at finite budgets and shows why it disappears from positive power-law leading asymptotics.

We also provide an end-to-end construction for the model. The endpoint subproblem is a known two-objective trace minimax problem; its solution uses scalar eigenvalue optimization and recovery inside a possible tied eigenspace. We use it within exact-real greedy interval covering, accelerate the repeated endpoint queries through reusable bounds, and attach solver-independent finite upper and lower witnesses to computed menus. An explicit separation shows why the scalar fully adaptive curve alone does not encode which finite menus can be realized.

The main structural results are the sharp model realizations in Theorem 3, the exactness criterion in Theorem 4, and the finite additional-preset bound in Theorem 6. Their ingredients include standard weighted scalarization, exposed faces, and interval covering. A reuse implementation and independently replayable continuous-price bounds complement this theory. We do not claim a new two-criterion duality theorem, interval greedy rule, eigenvalue-verification principle, or worst-case bit-complexity bound. We also do not claim a validated physical application: resource matrices in the experiments are synthetic.

Relation to existing methods. Regret-minimizing representative sets are established for finite databases, including efficient two-dimensional decision algorithms and interval-covering methods [4]. Succinct convex Pareto approximation already distinguishes feasible representatives from convexified points and identifies losses from restricting the representatives [8, 7]. Our candidate set is an implicit continuum of rank projectors with site-dependent costs, and the loss is additive over a specified price interval. Neither the finite-menu objective nor the use of interval covering is claimed to originate here.

Policy-count bounds in robust optimization also use finite convex representations [10, 3]. Recent bounds for nonlinear objectives require concavity in the uncertain parameter [13], whereas our regret is convex in price. Thus those dimension-based exactness bounds do not directly apply. Our bound instead compares a relaxed finite menu with a realizable finite menu at the same regret tolerance. Appendix D details this distinction and the limits of the contribution relative to convex Pareto approximation.

Tantipongpipat et al. [20] establish exact polynomial-time results for two-criterion dimensionality reduction. Shen et al. [19] develop scalar eigenvalue optimization for fair PCA, use Brent optimization, and explicitly discuss recovery at repeated eigenvalues. These are direct predecessors of our endpoint oracle, including the need for actual subspace recovery. Our timing references implement their scalar strategy in our outer menu construction; they are not runs of the authors' software.

Scalar convex-function approximation is also well developed. Rote's sandwich analysis [16] studies adaptive function/derivative evaluation; Imamoto and Tang [12] give a fixed-segment minimax construction for differentiable convex functions. Those problems permit scalar approximants defined from the function, whereas our affine pieces must come from a common task matrix and particular site costs. Proposition 1 demonstrates the distinction, rather than assuming it. Continuous covering and finite adaptability have broader precedents, including affine polyhedral families studied by Cardinal et al. [5]; their input class and complexity results differ from the rank-projector family here. Verified spectral computation is an established field [17]. Our verifier uses elementary congruence enclosures specifically to check menu witnesses. Approximation-rate results such as those of Gaubert et al. [9] are background, not a claimed new inverse-square law in this paper.

2 Problem and a structural separation

Let $A, C_1, \dots, C_m \in \mathbb{R}^{n\times n}$ be symmetric, $1 \le r \le n$, $m \ge 1$, and $B > 0$. For the resource interpretation, $A, C_L \succeq 0$ and $d_L \ge 0$. A preset is $P = UU^T$ with $U^TU = I_r$; the compact set of such projectors is denoted by $\mathcal{G}_r$. At price $p \in [0, B]$ and site L,

$$J_L(P, p) = -\operatorname{tr}(AP) + p\{\operatorname{tr}(C_L P) + d_L\}, \tag{1}$$

$$F_L(p) = \sum_{j=1}^{r} \lambda_j(-A + pC_L) + pd_L, \qquad F(p) = \min_L F_L(p), \tag{2}$$

Eigenvalues are in increasing order. The minimum trace principle gives $F(p) = \min_{P,L} J_L(P, p)$.

Write $\alpha(P) = -\operatorname{tr}(AP)$ and $c(P) = \min_L\{\operatorname{tr}(C_L P) + d_L\}$. Because $p \ge 0$ and A is shared, one cheapest site works for a fixed preset at every price. Its effective cost is the affine function

$$J(P, p) = \alpha(P) + pc(P), \qquad R(P, p) = J(P, p) - F(p). \tag{3}$$

F is continuous and concave, as an infimum of affine functions with bounded slopes. Thus $R(P, \cdot)$ is continuous, nonnegative, and convex. For a menu $Q = (P_1, \dots, P_K)$, define

$$E(Q) = \max_{p\in[0,B]} \min_i R(P_i, p), \qquad E_K = \min_{Q\in\mathcal{G}_r^K} E(Q). \tag{4}$$

Compactness and continuity ensure attainment. Repeating a projector permits menus with fewer than K distinct presets. A preset is an entire subspace; the budget is not a count of the rK basis vectors.

For $0 \le a < b \le B$, write

$$e(a, b) = \min_{P\in\mathcal{G}_r} \max\{R(P, a), R(P, b)\}, \qquad e(a, a) = 0.$$

Convexity makes this the optimal one-preset error on $[a, b]$. Section 4 gives a spectral construction for it.

Proposition 1. *The fully adaptive value function F does not determine E_K in this model. Moreover, restricting candidates to all globally price-optimal presets can strictly increase E_K, even when all choices at ties are allowed.*

Proof. Take $n = 2$, $r = 1$, $B = 1$, $A = \operatorname{diag}(1, 0)$, and zero overheads. Put $t = u_1^2 \in [0, 1]$. Model S has one site $C = \operatorname{diag}(2, 0)$, giving $J_S = -t + 2pt$. Model M has $C_1 = \operatorname{diag}(2, 2)$ and $C_2 = \operatorname{diag}(4, 0)$, giving $J_M = -t + p\min(2, 4t)$. Both yield $F(p) = \min(-1 + 2p, 0)$. In S the endpoint regrets are $1 - t$ and t; their minimized maximum is $1/2$ at $t = 1/2$. In M they are $1 - t$ and $\min(2 - t, 3t)$. On $[0, 1/2]$ the minimized maximum is $3/4$ at $t = 1/4$; on $[1/2, 1]$ its minimum is 1 at $t = 1$. Convexity makes these endpoint calculations sufficient on the whole domain.

In M the unique global optimum has $t = 1$ for $p < 1/2$ and $t = 0$ for $p > 1/2$. At $p = 1/2$ the regret is $\min(1 - t, t)$, which vanishes only for $t = 0, 1$. Hence the collection of all global optima contains only the two coordinate projectors; either alone has worst regret 1. The better $t = 1/4$ preset is supported by site 2 at $p = 1/4$, where that site's value is 0 but the global value is $-1/2$. It is never globally price-optimal. □

This is a model-specific illustration of nonconvex implementability, not a claim that scalar value functions generally preserve all feasible decisions. The unrestricted upper affine approximation

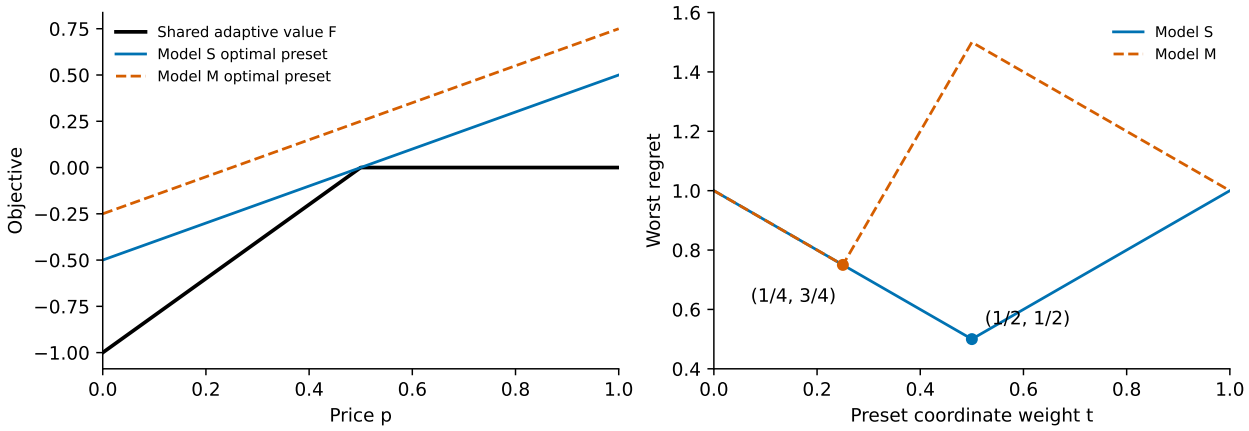


Figure 1: The same fully adaptive curve permits different finite menus. Left: the optimal realizable one-preset lines of Models S and M. Right: their endpoint maximum regrets as a function of $t = u_1^2$. The separation follows exactly from Proposition 1.

$p - 1/2$ achieves regret $1/2$ above this shared F, but cannot be implemented by a Model M preset. An algorithm given only F must return the same answer for S and M and therefore cannot solve both correctly. Sampling all sites is stronger than sampling global optima; our dictionary comparisons use all sites. Adding αI to A and a common βI to every cost matrix adds $-\alpha + \beta p$ to every rank-one objective and to F. Thus the separation persists with strictly positive definite matrices for $\alpha, \beta > 0$. Figure 1 depicts the example.

3 What the adaptive curve retains and loses

The separation in Proposition 1 raises quantitative questions. How far can a curve-only relaxation underestimate menu regret? When is it exact? Can additional globally optimal presets compensate for its failure? We answer these questions using the slopes available on the active spectral faces. The convex-geometric arguments apply more generally to compact families of affine functions; the spectral model supplies an explicit face test and finiteness of the exceptional prices.

3.1 Three feasible families

Identify an affine line with its coefficient pair and let $\mathcal{L} = \{(\alpha(P), c(P)) : P \in \mathcal{G}_r\}$, a compact subset of $\mathbb{R}^2$. Let $\mathcal{G} \subseteq \mathcal{G}_r$ contain every preset that attains $F(p)$ at some $p \in [0, B]$, including every choice at ties. This set is compact: it is the projection of the closed optimality relation in $\mathcal{G}_r \times [0, B]$. Write $E_K^{\mathcal{G}}$ for the menu optimum restricted to $\mathcal{G}$, and $\overline{E}_K$ for the optimum when each preset line can be any member of $\operatorname{conv} \mathcal{L}$. The latter is a mathematical relaxation; a convex combination of effective lines need not be realizable by a projector. The lower envelope remains F. Compactness ensures attainment of all three optima, and

$$\overline{E}_K \le E_K \le E_K^{\mathcal{G}}. \tag{5}$$

The geometric statements below hold for any nonempty compact coefficient set $\mathcal{L} \subset \mathbb{R}^2$, with presets interpreted as its elements and $F(p) = \min_{(\alpha,c)\in\mathcal{L}}(\alpha + pc)$. Convexity of $\mathcal{L}$ is not assumed. The spectral content is the realization of the sharp examples, the explicit active-face formula, and

finiteness of the exceptional set. In particular, these results do not assert a new general scalarization or convex-duality principle.

For $a < b$ define the curve-only endpoint value

$$D(a,b) = \max_{p\in[a,b]} \{F(p) - F(a) - (p-a)s\}, \qquad s = \frac{F(b)-F(a)}{b-a}. \tag{6}$$

For an interior price define the compact set of available optimal slopes

$$S(p) = \{c(P) : J(P,p) = F(p)\}. \tag{7}$$

The right and left derivatives of the compact affine envelope are $F'_+(p) = \min S(p)$ and $F'_-(p) = \max S(p)$, respectively. One way to see the right-derivative identity is to compare $F(p+h)$ with each minimizing line at p, then take a convergent subsequence of minimizing lines at $p+h$ as $h \downarrow 0$. The reverse inequality follows from optimality at p. The left identity follows in the same way. Thus the superdifferential is $[\min S(p), \max S(p)]$.

Lemma 2. *$D(a,b)$ is the exact endpoint minimax value for $\operatorname{conv}\mathcal{L}$. An attaining relaxed line is either a globally optimal preset line or a convex combination of two preset lines that are globally optimal at the same interior price.*

Proof. Every relaxed line lies above F. Its regret relative to the chord at any price is a convex combination of its two endpoint regrets, giving $D(a,b)$ as a lower bound. Maximize $F(p) - sp$ at p_*. If $a < p_* < b$, then $s \in [\min S(p_*), \max S(p_*)]$. Choose two optimal lines whose slopes bracket s and mix them to have slope s. They agree at p_*, so the mixed line has both endpoint regrets equal to $D(a,b)$. If $p_* = a$, select a minimizing line at a of smallest slope. The right-derivative optimality condition makes its slope at most s, and its regret at b is at most its regret at a. If $p_* = b$, use the largest minimizing slope and the left-derivative condition. The same argument uses the appropriate one-sided derivatives at 0 and B. □

The acceptable set of a relaxed line is again an interval. Consequently Theorem 8, with D in place of e, computes $\overline{E}_K$ from F alone in exact arithmetic. It computes a lower bound on the realizable menu problem, with the following sharp comparison. This is a statement about values and covering intervals: recovery of an actual projector still requires the model's active-face information.

Theorem 3. *For every $K \geq 1$,*

$$\overline{E}_K \leq E_K \leq E_K^{\mathcal{G}} \leq 2\overline{E}_K. \tag{8}$$

The constants in $E_K \leq 2\overline{E}_K$ and $E_K^{\mathcal{G}} \leq 2E_K$ cannot be decreased uniformly over the spectral models and budgets. Sharpness already occurs in the limit with $K = 1$.

Proof. An optimal relaxed menu supplies a partition into at most K segments, each served by one relaxed line at tolerance $\overline{E}_K$. Such a partition is obtained from its closed interval cover as in Theorem 8. On a segment $[a,b]$ choose any globally optimal preset at $q = (a+b)/2$. Its two nonnegative endpoint regrets sum to

$$2F(q) - F(a) - F(b) \leq 2\overline{E}_K,$$

because the covering relaxed line lies above $F(q)$ and its endpoint regrets are at most $\overline{E}_K$. Each endpoint regret is therefore at most $2\overline{E}_K$, and convexity extends this bound to the segment. One such preset per segment proves the last inequality. The other inequalities are inclusions of feasible families. The next example proves sharpness. □

A continuum of indistinguishable curves. Extend Model M by taking $C_1 = 2I$ and $C_2 = \mathrm{diag}(\kappa, 0)$ for $\kappa > 2$, retaining $A = \mathrm{diag}(1, 0)$, $r = 1$, $B = 1$, and zero overheads. For every κ,

$$F(p) = \min(-1 + 2p, 0), \qquad \overline{E}_1 = \tfrac{1}{2}, \quad E_1 = 1 - \kappa^{-1}, \quad E_1^{\mathcal{G}} = 1. \tag{9}$$

Indeed, with $t = u_1^2$ the endpoint regrets are $1 - t$ and $\min(2 - t, (\kappa - 1)t)$. For $t \le 2/\kappa$ their maximum is minimized at $t = 1/\kappa$, with value $1 - 1/\kappa$. For $t \ge 2/\kappa$ its minimum is 1. For $p > 0$ the only globally optimal projectors are the appropriate coordinate projectors, including both at $p = 1/2$; at $p = 0$ only the first coordinate is optimal. The relaxed value follows from (6). Hence

$$\frac{E_1}{\overline{E}_1} = 2 - \frac{2}{\kappa} \longrightarrow 2 \quad (\kappa \to \infty), \qquad \frac{E_1^{\mathcal{G}}}{E_1} = \frac{\kappa}{\kappa - 1} \longrightarrow 2 \quad (\kappa \downarrow 2).$$

Also $E_1^{\mathcal{G}} = 2\overline{E}_1$ throughout the family. The same curve realizes every value of E_1 strictly between $1/2$ and 1. Common positive definite shifts preserve the regrets. At $\kappa = 2$ the optimal face at $p = 1/2$ contains every projector, so the limiting claim does not assert continuity of the restricted feasible family.

The factor-two argument is the elementary two-objective weighted-sum bound applied to endpoint regrets. Weighted scalarization and limitations of supported solutions have a substantial approximation literature [2, 11]; we do not claim that principle as new. Here its role is to quantify the information loss in Proposition 1, with sharp examples inside the specified spectral model.

3.2 An exact criterion on active slopes

Call $p \in (0, B)$ a *slope-gap price* if $S(p) \neq [\min S(p), \max S(p)]$. This definition concerns missing slopes, not merely nondifferentiability: a corner can have every intermediate slope available from an optimal projector.

Theorem 4. *The identity $e(a, b) = D(a, b)$ holds for every $0 \le a < b \le B$ if and only if there are no slope-gap prices in $(0, B)$. In that case*

$$E_K = E_K^{\mathcal{G}} = \overline{E}_K \qquad (K \ge 1). \tag{10}$$

Proof. If every $S(p)$ fills its interval, the interior construction in Lemma 2 uses one actual globally optimal preset of slope s. Its boundary construction already uses one preset. Thus $e(a, b) = D(a, b)$ with a globally optimal realizing preset on every segment. Applying this to an optimal relaxed cover proves (10).

Conversely let s be a missing slope at $p_* \in (0, B)$. Compactness of $S(p_*)$ implies $l := F'_+(p_*) < s < u := F'_-(p_*)$. Fix $R > (u - s)/(s - l)$. For small $h > 0$ set $a = p_* - h$. The chord slope from a to p_* tends to $u > s$, whereas the chord slope from a to $p_* + Rh$ tends to $(u + Rl)/(1 + R) < s$. By continuity there is $b \in (p_*, p_* + Rh)$ for which the chord slope is exactly s. Concavity and $l < s < u$ make p_* the unique maximizer of $F(p) - sp$.

Suppose an actual preset attained endpoint error $D(a, b)$. At p_* the positive weighted average of its endpoint regrets equals $J(P, p_*) - F(p_*) + D(a, b) \ge D(a, b)$. Since each endpoint regret is at most $D(a, b)$, equality forces both regrets to equal $D(a, b)$ and $J(P, p_*) = F(p_*)$. Equal endpoint regrets force $c(P) = s$, contradicting $s \notin S(p_*)$. Attainment on the compact projector set therefore gives $e(a, b) > D(a, b)$. □

Computing the slope set from spectral faces. At an interior price let $\mathcal{A}(p) = \{L : F_L(p) = F(p)\}$. For each $L \in \mathcal{A}(p)$ use the cutoff-face construction in Theorem 7 to obtain its minimum and maximum costs $c_{L,-}(p)$ and $c_{L,+}(p)$. Then

$$S(p) = \bigcup_{L \in \mathcal{A}(p)} [c_{L,-}(p), c_{L,+}(p)]. \tag{11}$$

Every intermediate cost is realized by the explicit rotations in that theorem. Moreover, a minimizing projector of an active site must have that site among its cheapest sites: since $p > 0$, a strictly cheaper site would yield an objective below $F(p)$. Conversely, a globally optimal projector minimizes at a cheapest active site. This proves (11). The restriction $p > 0$ is essential to this argument; at zero price a tied site's raw cost need not be the projector's effective cost.

In particular, a single site has no slope-gap prices, even at eigenvalue collisions. Differentiability of F throughout $(0, B)$ is another sufficient condition, since then every $S(p)$ is a singleton. With multiple sites, one tests whether the union in (11) fills its interval. A jump in an optimizing subspace alone does not imply failure of the curve relaxation.

Remark 5. *The equivalence concerns relaxation exactness on* every subinterval. *It is not a converse criterion for* (10) *on a single fixed domain, or for the optimality of supported presets alone. For Model M on* $[1/2 - h, 1/2 + h]$, $0 < h < 1/4$, *direct endpoint calculation gives* $e = e^{\mathcal{G}} = 2h$ *but* $D = h$. *Here* $e^{\mathcal{G}}$ *restricts the endpoint problem to globally optimal presets. Thus that restriction can be exact even when convexification strictly improves the objective.*

3.3 A finite cost in additional presets

Theorem 6. *If there are* $q < \infty$ *slope-gap prices in* $(0, B)$, *then*

$$E^{\mathcal{G}}_{K+\min(K,q)} \le \overline{E}_K \le E_K \qquad (K \ge 1). \tag{12}$$

Without finiteness, $E^{\mathcal{G}}_{2K} \le \overline{E}_K$ *still holds. For the finite-dimensional spectral model,* q *is always finite.*

Proof. Partition an optimal relaxed cover into at most K segments and replace its line on each segment by an endpoint optimizer from Lemma 2. If a globally optimal preset realizes that line, retain it. Otherwise its slope lies in a gap of $S(p_*)$ at an interior maximizing price. Choose two globally optimal lines at p_* whose slopes bracket the required slope. Their lower envelope is no greater than their convex combination at every price. The two actual presets therefore cover this segment at its relaxed tolerance.

Each replacement that needs two presets can be charged to a slope-gap price strictly inside its segment. Distinct segment interiors are disjoint, so each price is charged at most once. There are at most $\min(K, q)$ extra presets. This is selection between two deterministic presets after observing the price, not a randomized projector.

For finiteness, apply Rellich's analytic eigenvalue parameterization to each affine symmetric pencil on an open interval containing $[0, B]$ [1, Theorem 1.1]. Each F_L is the minimum of the finitely many analytic functions obtained by summing any r of those eigenvalue branches and adding pd_L. Consequently F is the minimum of a finite family of analytic functions. Discard identical functions. Every pair then has finitely many zeros of its difference on the compact interval $[0, B]$. Outside their finite union, the minimizing analytic function is locally fixed, so F is differentiable. A slope gap requires nondifferentiability by (7), proving finiteness. No polynomial bound on q, or efficient exact enumeration of it, is asserted. □

Let $N_{\mathrm{rel}}(\epsilon)$, $N(\epsilon)$, and $N_{\mathcal{G}}(\epsilon)$ be the least menu sizes at tolerance $\epsilon > 0$ in the three families. They are finite, and

$$N_{\mathrm{rel}}(\epsilon) \le N(\epsilon) \le N_{\mathcal{G}}(\epsilon) \le N_{\mathrm{rel}}(\epsilon) + \min\{N_{\mathrm{rel}}(\epsilon), q\}. \tag{13}$$

Thus the extra supported-menu budget is bounded independently of the requested accuracy for each fixed spectral instance. The bound counts exceptional prices, rather than the number of missing slope intervals at one price. It can be conservative: different repairs may reuse a preset. Some overhead is necessary: the family in (9) has $q = 1$ and $E_1^{\mathcal{G}} = 1 > \overline{E}_1 = 1/2$, whereas its two coordinate presets give $E_2^{\mathcal{G}} = 0$. Thus the extra preset cannot be removed uniformly even for $q = K = 1$. We do not assert sharpness of the additive q term for every q and K.

For $K > q$, (12) also gives

$$\overline{E}_K \le E_K \le E_K^{\mathcal{G}} \le \overline{E}_{K-q}. \tag{14}$$

If a positive asymptotic law $\overline{E}_K \sim cK^{-\alpha}$ has been established for a particular curve, then both E_K and $E_K^{\mathcal{G}}$ have that same leading law. This transfers existing scalar approximation results, such as suitable one-sided smooth-curve laws [9]; it does not establish a new exponent or assert those regularity assumptions for every spectral instance. Piecewise affine curves can instead admit a finite zero-regret menu.

Relation to representative-set approximation. Minimum-size convex Pareto approximation already distinguishes feasible representatives from convex combinations [8]; the detailed treatment in [7, Sections 5.2–5.3] also exhibits loss from restricting representatives to extreme supported points. That phenomenon is not new here. Nor is replacement by supporting pieces: [14, Lemma 2] gives a bounded-cardinality tangent replacement for multiplicative scalar approximation. Our two-line replacement is the corresponding elementary exposed-face construction for additive regret. The refinement is to charge a replacement only at an interior price with an unavailable active slope; disjoint covering intervals charge each such price at most once. Combining this count with (11) and spectral finiteness gives the instance-wise additive overhead. It does not improve the computational-complexity guarantees of those predecessors. Appendix D makes the approximation criteria and the relation to robust policy-count bounds explicit.

4 Endpoint construction and global covering

For $a < b$, let $f_a = F(a)$, $f_b = F(b)$, and $s = (f_b - f_a)/(b - a)$. Define $e_L(a, b)$ as the best maximum endpoint regret at fixed site L. The following spectral reduction is a specialization of established two-objective trace minimax methods [20, 19]. We include recovery and the proof to specify exactly what our menu oracle needs.

Theorem 7. *For each site,*

$$e_L(a, b) = \max_{p \in [a,b]} \{F_L(p) - f_a - (p - a)s\}. \tag{15}$$

An actual rank-r projector attains this value, including at a repeated cutoff eigenvalue. The minimum endpoint regret with free site choice is $e(a, b) = \min_L e_L(a, b)$. Set $e(a, a) = 0$.

Proof. For any fixed projector, the convex combination of its endpoint regrets with weights $(b - p)/(b - a)$ and $(p - a)/(b - a)$ is its regret relative to the chord at p. Minimizing that combination gives the right side of (15) as a lower bound on the maximum endpoint regret.

Choose a scalar maximizer p_*. All minimizing projectors of $H = -A + p_* C_L$ contain its eigenspace strictly below $\lambda_r(H)$ and choose the remaining k dimensions within its cutoff eigenspace. Compress C_L to that tied eigenspace. If its eigenvalues are $\mu_1 \le \cdots \le \mu_q$, the minimum and maximum costs on the minimizing face are the fixed contribution plus d_L and respectively $\sum_{j=1}^{k} \mu_j$ and $\sum_{j=q-k+1}^{q} \mu_j$. Denote them by c_- and c_+.

Every intermediate cost is attainable by a pure projector. Retain the shared coordinates of the k-low and k-high eigenspaces of the compression. Pair the remaining low and high coordinates, replacing each low column e_i by $\sqrt{1-t}\, e_i + \sqrt{t}\, e_j$. Disjoint supports keep the columns orthonormal; their total cost is $(1-t)c_- + tc_+$. This is an explicit recovery construction, not random selection between two projectors. Tied-space recovery itself is prior art [19].

The right and left derivatives of the affine envelope F_L at p_* are c_- and c_+. At an interior maximum of (15), $c_- \le s \le c_+$. Select the face projector with cost s; its two endpoint regrets coincide with the dual value. At $p_* = a$, choose cost $c_- \le s$: the regret at b is no larger than at a. At $p_* = b$, choose cost $c_+ \ge s$ and reverse the argument. Equality follows in all cases. Finally, each projector has a single cheapest site at both endpoints, so minimizing over projectors and the finite site set commutes. □

The following interval-covering characterization is standard; compare [5, Section 4.2]. We include its compactness argument because the preset set here is an implicit continuum.

Theorem 8. *For $\epsilon \ge 0$ define*

$$T_\epsilon(x) = \max\{y \in [x, B] : e(x, y) \le \epsilon\}. \tag{16}$$

Starting at $g_0 = 0$, set $g_j = T_\epsilon(g_{j-1})$, with $T_\epsilon(B) = B$. Then $E_K \le \epsilon$ if and only if $g_K = B$.

Proof. The acceptable prices for a preset form a closed interval, so two acceptable endpoints are equivalent to acceptability on their whole interval. A maximizing y and its preset exist by compactness. Further, T_ϵ is nondecreasing: a preset covering $[x, T_\epsilon(x)]$ also covers $[x', T_\epsilon(x)]$ if $x \le x' \le T_\epsilon(x)$; otherwise $T_\epsilon(x') \ge x' > T_\epsilon(x)$.

Greedy success constructs a menu. Conversely, a feasible finite menu gives a closed interval cover, from which a chain of at most K overlapping intervals connects 0 to B. If its reachable endpoints are x_j, then $x_j \le T_\epsilon(x_{j-1})$. Induction using monotonicity gives $x_j \le g_j$, forcing $g_K = B$. □

For $\epsilon > 0$ progress is uniformly positive unless all preset slopes coincide. If they lie in $[c_{\min}, c_{\max}]$, a price-optimal preset at x has regret at most $(c_{\max} - c_{\min})|y - x|$ at y. Thus the covering needs finitely many steps for positive tolerance. When all slopes coincide, one optimal preset suffices everywhere. The theorem supports a monotone outer search for E_K, but is an exact-real characterization, not a complexity guarantee for finite-precision nested searches.

Numerical construction. For each candidate tolerance, the implementation makes at most K farthest endpoint searches. Each uses bisection in price and Theorem 7 to test the interval, returning a realized preset. An outer bisection searches the regret tolerance. For $K = 1$ it solves only the full-domain endpoint problem. At a site, the scalar objective in (15) is concave; slope bisection and face recovery give primal and dual estimates. A site is feasible as soon as an actual preset upper estimate is at most ϵ; exclusion requires lower estimates above ϵ at every site. Near eigenvalues are grouped only to propose a basis, whose actual primal objective is then evaluated. Unresolved oracle gaps raise an error. The numerical outer-search lower estimate is not a certified lower bound: ambiguous finite-precision decisions and finite price resolution are handled by separate witnesses below.

Computational work. Let N_ϵ and N_p be the numbers of outer tolerance and price bisections, and let T bound the spectral evaluations needed by a completed site endpoint solve, including recovery. A conservative uncached work count is $O(N_\epsilon K(1+N_p)mTn^3)$ with dense eigensolvers. Tolerance bisection gives logarithmic N_ϵ, N_p in the respective search resolutions, but a uniform accuracy-dependent bound for T near arbitrary collisions is not asserted. The implementation caps it and reports unresolved gaps. Saved-witness verification uses $O(Kmn^3)$ dense arithmetic operations, independently of the number of outer search queries. Its constants include explicit interval products; verification is timed separately from optimization.

5 Reuse and independently checkable bounds

The outer construction produces many related endpoint queries. A stored site-preset line $\alpha + pc_L$ bounds $e_L(a,b)$ from above by

$$\max\{\alpha + ac_L - F(a), \alpha + bc_L - F(b)\}. \tag{17}$$

A stored value $F_L(q)$ gives a lower bound $F_L(q) - F(a) - (q-a)s$ whenever $q \in [a,b]$. Minimizing the cached primal bounds and maximizing eligible dual bounds can settle a feasibility query without a new eigendecomposition.

A minimizing preset at q also gives a supergradient c_L of F_L. If $c_L > s$, the supergradient inequality rules out scalar maximizers strictly to the left of q; if $c_L < s$, it rules out those strictly to its right. The cache therefore brackets subsequent scalar searches. These are standard supporting-line principles. We compare slope bisection with Brent both without this reuse and with increasingly complete reuse. Inverted numerical brackets are discarded. Full eigensystems use a bounded cache of about 32 MiB; supported basis records require additional memory, which is reported.

Proposition 9. *Suppose K presets and a contiguous partition of $[0,B]$ have endpoint regret at most U, assigning one preset to each segment. Suppose also that $K+1$ increasing prices $x_0 < \cdots < x_K$ satisfy, for every adjacent pair,*

$$\min_L\{F_L(q_{i,L}) - \mathrm{chord}_{x_i,x_{i+1}} F(q_{i,L})\} > \ell, \qquad q_{i,L} \in [x_i, x_{i+1}]. \tag{18}$$

Then $\ell < E_K \le U$.

Proof. Convexity extends each segment's endpoint upper bounds to the full segment. The weak-dual part of Theorem 7 shows that no preset can serve an adjacent witness pair at tolerance ℓ. Nor can it serve a nonadjacent pair, since its acceptable set is an interval. Hence each preset serves at most one witness, and K presets cannot serve $K+1$ witnesses. □

The verifier is separate from the optimizer. Stored binary64 matrices mean their exact real values. A stored basis U means its exact polar-normalized subspace $Q = U(U^TU)^{-1/2}$; it need not be exactly orthonormal as stored. Outward interval arithmetic encloses its objective traces and the branch spectral sums. The verifier checks segment adjacency, endpoints, witness ordering, all site inequalities, and the final gap. Appendix B gives the enclosure argument and arithmetic contract. The checked eigensolver only proposes a basis for an enclosure; neither its nominal precision nor the menu solver's stopping test is trusted as a certificate.

For ordinary numerical evaluation, all pairwise intersections of the menu's effective affine lines, together with $0, B$, suffice to find its continuous maximum regret. Between crossings the lowest line is fixed and its regret is convex. The interval verifier avoids the need to certify those computed crossings by checking the saved covering segments instead.

6 Computational evidence in brief

The experiments test the construction and its certificates; they do not establish the structural theorems or a physical resource model. Appendix A retains the protocols, all comparison tables, stress cases, and failed development attempts. The original three dense studies contain 402 timed runs and 134 saved-menu checks. Against hybrid Brent with the same cached primal/dual decisions, the median ratio of reference time to reuse time is 1.367 across 20 declared cells; its range is $[1.255, 1.866]$. Specialized two-dimensional formulas are faster than the general construction. These are implementation comparisons, not a new worst-case complexity result.

The dense studies and dictionary references include 54 independently replayed global certificate artifacts of width 10^{-5}, counting repeated controls. Separately, the sharp family (9) has 12 outward checks matching its exact formula to below 10^{-10}. The other structural diagnostics are numerical checks, not proofs.

The sparse confirmation uses 72 timed runs in 12 paired cells. Dense partial eigensolves are faster at dimensions 128 and 256; the sparse backend wins at dimension 512 with cell ratios 1.39–1.90. Larger scaling cases are single observations. Independent banded verification certifies all 14 saved sparse grid menus, including dimension 4096, with gaps below 1.01×10^{-5}. Four of those grids also have earlier dense certificates: these counts describe artifacts, not additional independent instances. Banded verification exploits limited bandwidth, rejects unresolved pivots, and is timed separately from optimization. No performance claim for arbitrary sparse matrices or GPU execution is made.

7 Scope and conclusions

The useful structure is a shared reward, a scalar nonnegative price, and quadratic site costs for a fixed-rank subspace. It yields an exact-real finite-menu characterization and permits finite independent witnesses. The scalar adaptive curve determines a relaxation that can underestimate realizable regret by a factor approaching two. Missing optimal slopes characterize failure of this relaxation on subintervals. They occur at only finitely many prices in each fixed spectral instance, and their number bounds the additional globally optimal presets needed to repair a relaxed menu. Thus finite-budget distinctions coexist with agreement of positive power-law leading asymptotics. The active-face test identifies cases where curve approximation is already exact, including every single-site instance. Site-specific information is needed to resolve the general realizability constraint, not in every special case. Repeated endpoint queries benefit from reuse on the reported dense families, although the advantage narrows against a fully warmed Brent reference and reverses against specialized two-dimensional formulas.

The sparse extension tests partial eigensolves at matched accuracy and extends independent outward verification to banded inputs. Its scope is specific graph-plus-diagonal families; it does not establish performance for arbitrary sparse matrices, very large dimensions, many resource parameters, site-dependent task rewards, negative prices, or externally measured resource models. Verified grid bounds exploit their bandwidth; arbitrary sparse graphs may have excessive fill or inconclusive pivots. Additional basis storage can grow even when the spectrum cache is bounded. Numerical convergence near spectral collisions is not a bit-complexity guarantee. The central contribution is consequently a structural characterization of convexification for finite spectral menus, supported by a construction with checkable solutions. The extra-preset count is instance-dependent; neither an efficient exact enumeration of exceptional prices nor a uniform small bound on their number is supplied.

Statements and declarations

Data and code availability. The accompanying artifact contains source code, frozen protocols, raw runs, saved menus, witness files, and solver-independent verification commands. It also contains the source for regenerating the tables and figure. A public repository or archival identifier has not yet been assigned.

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A Computational protocols and complete results

A.1 Design and reproducibility

The original three comparisons use our Python implementations with dense full eigendecompositions, one BLAS thread, and fresh model/cache state per timed run. Random matrices are $XX^T/n + 0.1I$ with independent standard-normal entries in X; site overheads are independent uniform draws in $[0, 0.5]$, and $B = 2$. Shapes (n, r, m) are $(8, 1, 1)$, $(16, 4, 3)$, $(32, 8, 6)$, $(64, 4, 3)$, and $(64, 16, 3)$. Each family uses two seeds and budgets $K = 2, 4$. The two-dimensional mirror model is a separate development control. We use objective, price, and endpoint tolerances 2×10^{-6}, 2×10^{-8}, and 10^{-9}, respectively. These are numerical search settings, distinct from the final verified gaps.

The initial confirmation uses seeds 5100/5101, two methods, and three repetitions (132 runs). The stronger-reference study uses new seeds 5400/5401, three methods, and three cyclically ordered repetitions (198 runs). Plans and source hashes were saved before their respective executions; these are local records, not externally registered protocols. Model generation and file saving are excluded from timings; cache building, search, recovery, and numerical continuous maximum evaluation are included. Certificate generation and independent checking are separate costs. Results use ratios of per-cell median times, excluding the mirror control from the primary summaries. No sample was removed for unfavorable timing. The recorded environment is Python 3.12.10, NumPy 2.5.2, SciPy 1.18.1, on Windows 11 with an AMD Ryzen 7 5800H processor. Hardware and library details accompany the artifact.

Independent menu checks use a separate eigenvalue evaluation, verify basis orthogonality and segment partitions, and evaluate every affine crossing. A 101-point price grid is only a secondary consistency check. The initial and stronger studies have 44 and 66 independently checked saved menus, respectively. All paired numerical losses agree within 5×10^{-6}.

Table 1: Stronger-reference study. Each row aggregates four cells (two seeds, two budgets). Ratios are reference seconds divided by proposed seconds; larger values favor reuse with slope bisection. Memory is retained spectrum and sample storage, not process peak RSS.

n	r	Sites	Warm Brent	Hybrid Brent	Max. MiB
8	1	1	1.920	1.810	1.7
16	4	3	1.917	1.571	14.0
32	8	6	1.630	1.330	52.4
64	4	3	1.456	1.330	42.8
64	16	3	1.470	1.330	79.9

A.2 Stronger scalar references

The first Brent reference includes endpoint-aware early decisions and actual primal recovery, but does not reuse bounds across different endpoint queries. Its median time divided by the reuse method's time is 2.810 on the 20 new-seed cells, with range $[2.112, 5.934]$. This is not the headline gain: warming the reference makes the comparison substantially closer.

The stronger study compares cached slope bisection with (i) Brent bracketed by cached slopes, and (ii) hybrid Brent using both those brackets and the same cached primal/dual decisions. Both Brent references retain a checked recovery fallback if their parameter convergence does not close the objective gap; fallback time is charged. Table 1 reports the results. The median ratios are 1.630 for bracket-warmed Brent and 1.367 for hybrid Brent, with ranges $[1.375, 2.033]$ and $[1.255, 1.866]$. No reuse slowdown occurs in these 20 declared $K = 2, 4$ cells. This supports a modest implementation advantage in this regime, not universal superiority of bisection over Brent or a claim about prior authors' measured performance.

A separate ablation holds slope bisection fixed and removes the supporting-line, dual-sample, and bracket reuse while retaining the same cache for identical spectral queries. It uses seed 5500 in each shape, $K = 2, 4$, three alternating repetitions, and the mirror control (72 runs, 24 checked saved menus). All numerical loss differences are below 5×10^{-6}. Over the ten new-seed cells, the plain-to-reuse median time ratio is 4.220, with range $[2.756, 7.378]$. This isolates the combined reuse mechanism; it does not isolate each cached-bound type, and it is a weaker comparison than hybrid Brent.

A.3 Dictionaries and special cases

The dictionary comparison selects presets over the continuous price domain, rather than judging them only on a grid. Uniform dictionaries contain every site's minimizing basis at 129, 513, 2049, or 8193 prices. Selection uses the same interval-covering geometry restricted to those candidates. An adaptive heuristic starts at three prices per site, refines around selected samples, and adds all-site minimizers at the menu's current worst price, for at most 20 rounds. It has no additional targeted tied-space recovery. This is our adaptive reference, not a reproduction of a published strongest possible sampling algorithm.

Ten cases include the mirror model with $K = 1, 2, 4$, three new random models (seed 5200) with $K = 2, 4$, and one commuting collision control with $K = 1$. Stopping uses a generous oracle-assisted target: the proposed menu's numerical loss plus 10^{-5}. The dictionary is given that target, but not the proposed preset directions. All candidate construction, refinement, and repeated selection work is charged cumulatively. Final endpoint upper bounds are verified independently. The reference

Table 2: Dictionary screen: cumulative seconds and target attainment at the stopping stage or cap. "No" is failure to reach the accuracy target, not a software failure. Verification time is excluded for all methods.

Case	K	Reuse (s)	Uniform (s)	Met?	Adaptive (s)	Met?
Mirror	1	0.005	0.766	No	0.015	Yes
Mirror	2	0.174	0.928	No	0.491	Yes
Mirror	4	0.392	1.258	Yes	1.039	Yes
n16 r4 s3	2	0.321	2.935	No	1.637	No
n16 r4 s3	4	0.872	3.924	No	3.584	Yes
n32 r8 s6	2	0.837	13.613	No	9.358	Yes
n32 r8 s6	4	2.277	17.129	No	21.652	Yes
n64 r4 s3	2	1.209	18.032	No	17.157	No
n64 r4 s3	4	2.691	22.925	No	44.356	No
Collision	1	0.001	0.379	No	0.019	No

Table 3: Specialized mirror solver versus general reuse, median seconds from three repetitions. All numerical loss differences are below 5×10^{-6}.

K	Analytic (s)	General reuse (s)
1	0.0015	0.0042
2	0.0067	0.1652
4	0.0128	0.3772

menus also have global certificates of width 10^{-5}. Comparing the verified upper bounds with those global lower bounds gives gaps below 1.94×10^{-5} for every dictionary menu that meets its target.

Uniform dictionaries meet their target in 1 of 10 cases within the cap; the adaptive heuristic meets it in 6 of 10. Table 2 reports every case, including failures to attain the target. These are single-run descriptive timings. Coarser dictionaries can be faster when less accuracy is required; these capped tests do not refute sampling with other refinement rules or collision recovery. The separate commuting control has one-preset loss $1/2$, while sampled coordinate eigenbases give 1. Unlike Proposition 1, that control depends on which basis the eigensolver chooses at a tie and does not exclude all choices there.

The original mirror example has $A = \left(\begin{smallmatrix} 1 & 0.6 \\ 0.6 & 2 \end{smallmatrix}\right)$, $C_1 = \operatorname{diag}(0.2, 3)$, $C_2 = \operatorname{diag}(3, 0.2)$, and overheads $0, 1$. It admits an independent analytic angular endpoint construction. At matched stopping resolution it is much faster than the general method (Table 3). This limits the practical recommendation: use specialized formulas when available. Development one-preset tests also favor Brent on several smooth examples; no $K = 1$ speed advantage is claimed.

A.4 Checks of the structural results

The script `check_structural_theory.py` checks six values of κ in (9), each with and without a positive definite shift. All 12 cases have optimizer-independent outward upper/lower bounds of width below 10^{-10} matching the exact formula. A further 54 numerical endpoint checks use dimensions $2, 6, 8$, ranks $1, 2, 4$, one or three sites, three seeds, and three intervals per instance. They check the factor-two bound and single-site equality; these are diagnostics, not interval proofs. An exact one-site crossing and three short Model M intervals check the distinction in the remark following Theorem 4.

A three-coordinate, three-site example has $A = \mathrm{diag}(2, 1, 0)$, $C_1 = \mathrm{diag}(3, 8, 8)$, $C_2 = \mathrm{diag}(8, 1, 8)$, $C_3 = \mathrm{diag}(8, 8, 0)$, zero overheads, $r = 1$, and $B = 2$. Its adaptive curve is $\min(-2 + 3p, -1 + p, 0)$, with slope gaps at $p = 1/2$ and $p = 1$. The relaxed two-cover splits at $2/3$, with endpoint error $1/4$ on both segments. Direct selection from the three globally optimal coordinate presets gives two-menu loss $1/3$ and three-menu loss zero. The numerical unrestricted two-menu construction also gives $1/3$; this last numerical value is not used as a proof. These examples illustrate conservative repair counts and the distinction between convexification and restriction to supported presets.

A.5 Verification, stress cases, and retained failures

The 22 proposed menus in each confirmation study and the 10 dictionary reference menus have independently replayed global upper/lower bounds of width 10^{-5}. These are 54 saved certificate artifacts, including repeated mirror controls, not 54 distinct random problems. They prove bounds for their exact binary64 instances under Appendix B's contract. Certificate generation may fail and is not asserted to succeed for all inputs.

We checked 24 site SDP solves on 12 instances with CVXPY and Clarabel (agreement within 2×10^{-7}); some SDP statuses include the recorded `optimal_inaccurate` status. Twelve orthogonal rotation checks agree within 2×10^{-9}. Twenty analytic stress cases cover exact and near crossings, scales 10^{-6} through 10^{6}, an eightfold tie with several ranks including full rank, and dimension one. Interval primitive checks compare 600 scalar cases and 87 matrix entries with exact fractions, 84 eigenvalues with 80-digit arithmetic, and 18 perturbed-basis polar objectives with high-precision calculations. These checks support the implementation; they are not formal software verification.

Two development failures are retained. An initial certificate generator mistook a cached primal basis's source price for a dual witness price; one out-of-range proposal was rejected by the verifier. A revised generator proposes a separate in-range price for each site and pair, and all 22 cases then pass. A dictionary run initially stopped on serialization of a NumPy Boolean; the corrected version changes serialization only. Neither incident is hidden by deleting its result directory.

A.6 Partial spectra and sparse computation

The preceding studies use full dense spectra. A separate extension stores the input matrices in compressed sparse row format and computes a partial spectrum through ARPACK, using SciPy's `eigsh` interface [18]. The initial request is $\min(n, \max(r + 8, 2r))$ vectors; it expands when the returned boundary is too close to the rank cutoff. QR and Rayleigh–Ritz refinement, residual checks, and a 32 MiB spectrum cache accompany the same reusable endpoint search. Dense fallbacks are explicitly recorded and disabled above $n = 1024$. This is an implementation of established sparse spectral methods, not a new eigensolver.

The comparison uses a dense *partial*-spectrum LAPACK backend with the same outer search and recovery. Both receive identical synthetic positive-semidefinite matrices: independently weighted graph Laplacians plus positive diagonal terms. Edge weights lie in $[0.05, 0.25]$, reward diagonals in $[0.2, 2]$, resource diagonals in $[0.1, 1.5]$, and overheads in $[0, 0.15]$. The graph is a row-major grid of width $\lfloor \sqrt{n} \rfloor$, possibly with an incomplete final row. These are computational test families without a calibrated physical interpretation.

The initial sparse study stopped after 18 of 72 planned runs: both backends failed all three repetitions of one rank-two case at $n = 128$. A narrow avoided eigenvalue crossing caused price bisection to reach binary64 resolution before meeting the 10^{-9} endpoint tolerance. The revised implementation rotates the adjacent cutoff vectors to match the required chord slope, evaluates the original primal objective, and accepts only under the unchanged gap or decision criterion. The

Table 4: Sparse confirmation. Times are medians of three runs in seconds; the ratio is dense partial divided by CSR/ARPACK. A ratio above one favors the sparse backend. Each row has two sites and two presets.

n	r	Seed	Dense partial	CSR/ARPACK	Ratio
128	2	6400	1.60	6.63	0.24
128	2	6401	1.24	5.01	0.25
128	8	6400	2.08	7.67	0.27
128	8	6401	2.01	7.16	0.28
256	2	6400	4.97	9.72	0.51
256	2	6401	4.24	6.99	0.61
256	8	6400	6.22	10.71	0.58
256	8	6401	6.11	11.16	0.55
512	2	6400	23.64	12.43	1.90
512	2	6401	22.08	11.75	1.88
512	8	6400	26.02	14.25	1.83
512	8	6401	40.09	28.81	1.39

six failures and their source snapshot are retained. This recovery is applied to both backends; the failed study's seeds are thereafter treated as development data.

A new protocol uses fresh seeds 6400 and 6401, $n = 128, 256, 512$, $r = 2, 8$, two sites, and $K = 2$. Three repetitions and two backends give 72 timed runs in 12 paired cells, with alternating method order, one BLAS thread, and the preceding tolerances. Every saved menu is checked with independent full dense eigenvalue evaluation at all affine-line crossings and covering-segment endpoints. Those numerical cross-checks are distinct from outward certificates. Table 4 reports per-cell median times; smaller instances need not benefit from sparse computation.

All 72 confirmation runs completed, and all 24 saved menus passed the independent numerical check. The median dense-partial/CSR runtime ratio across the 12 cells is 0.564. Its per-dimension medians are 0.259, 0.564, and 1.853 for $n = 128, 256, 512$, respectively. Thus the small-instance overhead must be weighed against the measured larger-instance behavior, rather than reporting only a favorable size. The maximum difference between reported and independently evaluated losses over confirmation and scaling menus is 2.5×10^{-14}. The sparse runs in these two studies used 0 dense fallbacks.

Table 5 extends the numerical screen to $n = 4096$. For the single $n = 1024$ grid, the dense-partial/CSR ratio is 7.59; this one observation is not a repeated speedup estimate. Independent ordered tridiagonal, full dense, and banded LAPACK calculations check path, $n = 1024$ grid, and $n = 4096$ grid menus, respectively. The banded check uses the known grid bandwidth, not the ARPACK proposals.

Outward verification and standalone replay also succeed for the four $n = 128$ CSR grid menus and all four path menus in the scaling screen, each with a global bound width 10^{-5} to displayed precision. The grid certificates use dense arithmetic; the path certificates use the tridiagonal verifier. The banded verification study below extends certification to the larger grids. Recorded array storage includes all cached preset bases, which can exceed the spectrum-cache budget; it excludes transient solver workspaces and Python overhead and is not a peak-memory measurement.

Partial eigenpair residuals do not establish extremal ordering or complete cutoff multiplicity. The grid results therefore use separate ordered spectral checks, with outward dense verification on selected tractable cases. For paths, an outward Sturm-count verifier certifies ordered eigenvalue sums without a dense matrix, using classical tridiagonal inertia[6]. Appendix C gives the argument. The

Table 5: Descriptive scaling screen: one solve per method and case, $r = 2$, two sites, $K = 2$. These are single observations, not repeated confirmation estimates. Retained memory counts input, spectrum-cache peak, and final basis-sample arrays, not peak process RSS.

Graph	n	Backend	Seconds	Retained MiB
Path	1024	CSR/ARPACK	17.09	60.6
Path	1024	Tridiagonal	5.03	54.5
Path	4096	CSR/ARPACK	64.34	140.8
Path	4096	Tridiagonal	23.98	140.8
Grid	1024	Dense partial	134.49	83.6
Grid	1024	CSR/ARPACK	17.73	59.8
Grid	4096	CSR/ARPACK	189.80	356.3

Table 6: Independently replayed outward certificates for CSR grid menus. Decimal counts record successful higher-precision inertia retries; the displayed gap column is rounded. All unrounded gaps are below 1.01×10^{-5}.

n	Menus	Half-bandwidth	Decimal counts	Max. gap / 10^{-5}
128	4	11	18	1.000029
256	4	16	26	1.000031
512	4	22	13	1.000031
1024	1	32	0	1.000008
4096	1	64	0	1.000008

descriptive larger-instance screen also compares paths against a specialized tridiagonal LAPACK solver; general CSR performance should not be inferred from beating an unnecessarily dense path implementation.

Outward verification of every CSR grid menu. A further verification study certifies all 12 CSR confirmation menus and both CSR grid scaling menus, through $n = 4096$, using interval LDL^T elimination in band storage. All 14 certificates pass standalone replay, with global gaps below 1.01×10^{-5} (Table 6). This strengthens the preceding numerical checks without changing any optimizer, saved menu, or isolated runtime measurement.

An initial binary64 interval implementation rejected a rank-eight witness because its eigenvalue brackets became too wide to prove exclusion. That attempt is retained. The revised verifier retries unresolved counts at $n \le 512$ with directed 50-digit decimal intervals; 57 counts use this path. Both larger grid cases need only outward binary64 arithmetic. The final checker independently encloses ordered eigenvalue sums and actual polar-orthonormalized projectors; it imports no menu solver. Six CPU processes check independent price/site pairs. These parallel verification times are separate from the isolated optimization study.

For half-bandwidth b, each unpivoted inertia count uses $O(nb^2)$ work and $O(nb+b^2)$ storage. The grid has $b = \lfloor \sqrt{n} \rfloor$; at $n = 4096$, packed input bands occupy about 6.1 MiB and an interval factor about 4.1 MiB per worker, excluding temporary arrays and process overhead. This is an advantage of band structure, not a guarantee for arbitrary sparse graphs. Ambiguous pivots cause rejection; high bandwidth can erase the storage benefit. Appendix C gives the factorization argument and arithmetic assumptions.

B Outward bounds for eigenvalues and actual subspaces

The arithmetic contract interprets each binary64 input as an exact real number. Elementary $+, -, \times, /$ operations are widened using `nextafter`; matrix products use explicit interval sums, not uncontrolled BLAS reductions. Correctly rounded IEEE-754 binary64 elementary operations and gradual underflow are assumed. Nonfinite intermediates or failed rank checks preclude certification. There is no formal proof-assistant verification or independent external audit of the code.

For an exact symmetric $H = -A + pC_L$, an approximate eigensystem V, D is only a proposal, with D sorted. Interval products enclose $G = V^T V$ and $M = V^T H V$. Let $\eta \geq \|G - I\|_2 < 1$ and $\rho \geq \|M - D\|_2$, obtained from absolute row-sum bounds for the enclosed symmetric matrices. The generalized Rayleigh quotient differs from the Rayleigh quotient of D by at most

$$\delta = \frac{\rho + \|D\|_2 \eta}{1 - \eta}. \tag{19}$$

Indeed $x^T G x \geq (1 - \eta)\|x\|^2$; subtracting the two quotients and bounding the residual numerator gives (19) uniformly in x. Since $\eta < 1$, V is invertible. Generalized Courant–Fischer then yields $|\lambda_j(H) - D_{jj}| \leq \delta$ for every j. Summing the first r enclosures and adding an interval for pd_L encloses F_L. Taking the minimum of all lower endpoints and the minimum of all upper endpoints encloses F.

For a stored preset basis U, let $\eta_U \geq \|U^T U - I\|_2 < 1$. Its polar basis $Q = U(U^T U)^{-1/2}$ exists and

$$\|UU^T - QQ^T\|_* = \sum_{j=1}^{r} |\sigma_j(U)^2 - 1| \leq r\eta_U. \tag{20}$$

Therefore an interval for $\mathrm{tr}(U^T H U)$, enlarged by $r\|H\|_2\eta_U$, contains the objective of the exact realizable projector QQ^T. Adding pd_L and subtracting the enclosure for F gives endpoint regret bounds. On each saved segment the verifier chooses one fixed site and uses its two endpoint bounds; convexity extends them to the interval. Lower witnesses use outward arithmetic for every quantity in (18). This checks the final optimization gap without running or importing the menu optimizer.

C Tridiagonal and banded outward verification

For tridiagonal $H = -A + pC_L$, write its diagonal as a_i and off-diagonal as b_i. At a real shift x, the recurrence

$$t_1 = a_1 - x, \qquad t_i = a_i - x - b_{i-1}^2 / t_{i-1}$$

gives an LDL^T factorization of $H - xI$ whenever all pivots are nonzero. By Sylvester's inertia law, the number of negative pivots equals the number of eigenvalues strictly below x; this is the classical Sturm count used in eigenvalue bisection[6]. Our verifier encloses every pivot by outward interval operations and accepts a count only if each interval is finite and excludes zero. It rejects unresolved signs.

An approximate eigenvalue merely proposes a bracket. If certified counts at l, h satisfy $N(l) \leq j-1$ and $N(h) \geq j$, then $\lambda_j(H) \in [l, h]$. Summing the first r enclosures and adding pd_L encloses $F_L(p)$. This argument permits multiplicities and does not rely on the ordering of approximate Ritz pairs. For a preset basis, evaluate

$$\mathrm{tr}(U^T H U) = \sum_{k=1}^{r} \left(\sum_i a_i U_{ik}^2 + 2 \sum_i b_i U_{ik} U_{i+1,k} \right)$$

by interval sums, then apply the polar-basis correction from Appendix B. An outward row-sum bound controls $\|H\|_2$. The same saved segment and packing witnesses give global menu bounds.

No dense $n \times n$ array is required. Each successful Sturm count uses $O(n)$ work, and the preset Gram calculation uses $O(nr^2)$ work. These statements do not bound the number of bracket expansions or menu search iterations. The implementation assumes the same elementary arithmetic contract as the dense verifier and has not received an external code audit.

For a symmetric matrix of half-bandwidth b, the same inertia argument applies to unpivoted band LDL^T elimination. At each step, if the pivot interval excludes zero, interval arithmetic encloses the Schur update $S_{ij} - v_i v_j / d$. Induction encloses the exact factorization, with no fill outside the full band. Definite signs of all pivots certify the inertia. A singular leading principal submatrix can prevent this calculation even for a nonsingular matrix; the checker rejects such a case. One successful count uses $O(nb^2)$ arithmetic operations and $O(nb + b^2)$ storage. A banded eigensolver only proposes brackets, which are checked by counts.

The trace calculation extends to every stored off-diagonal band, and the same polar correction gives an objective enclosure for an actual projector. Outward pairwise addition trees are valid by induction on their nodes. Independent branch-price checks can run in CPU processes without changing these arguments.

Unresolved counts at $n \le 512$ may be retried with directed 50-digit decimal intervals. Exact conversion of each binary64 input and separately downward/upward-rounded elementary operations enclose the same real matrix. This path additionally assumes the correct directed rounding of Python's `Decimal` implementation[15]. Nonfinite or zero-containing pivots still cause rejection, and decimal underflow is trapped. No automatic decimal fallback is used for larger inputs. These precision and bandwidth limits are implementation choices, not unconditional convergence guarantees.

D Comparison with representative-set and policy-count results

The affine coefficient representation clarifies both the connection to prior work and the precise scope of the structural results. This section does not claim that additive regret creates an unrelated optimization problem: its supporting-line geometry is shared with established representative-set methods.

Convex Pareto approximation. For a finite coefficient menu $Q \subset \mathcal{L}$,

$$\min_{z \in Q} \langle (1, p), z \rangle = \min_{z \in \operatorname{conv} Q} \langle (1, p), z \rangle.$$

Thus taking a convex hull *after selecting* the menu does not change its value at any price. Allowing its K selected coefficient points to come from $\operatorname{conv} \mathcal{L}$, however, can change the optimum. This is the distinction between the representative's feasibility and convex evaluation of the selected set. In particular, use of convex combinations in a Pareto-approximation definition does not itself distinguish that literature from finite-menu evaluation.

Diakonikolas' detailed treatment [7, Chapter 5], following [8], compares representatives selected from a finite set with representatives allowed in its convex hull. Its Section 5.2.2 explicitly recognizes the usefulness of points away from the lower convex envelope. Theorems 5.3.3 and 5.3.5 concern multiplicative approximation and cardinality with a weighted-optimization oracle; Remark 5.3.6 distinguishes objective-space points from recoverable decisions. Our separation example is a spectral realization of an established obstruction. Our contribution concerns the all-subinterval exactness criterion, pure spectral face recovery, and a repair count localized to missing-slope prices, with every

tied optimal preset retained in $\mathcal{G}$. Merely changing multiplicative to additive error would not by itself establish a substantive contribution.

Scalarization and replacement. Weighted-sum approximation factors and supported-solution limitations are established in [2, 11]. The factor-two proof here is their elementary endpoint-regret counterpart; the sharp examples place that loss inside the specified spectral model. Similarly, Magnanti and Stratila [14, Lemma 2] replace multiplicative piecewise-linear approximants with tangent pieces. Our refinement preserves an additive tolerance and counts the particular prices at which a realizable supporting line is unavailable. Rellich's theorem supplies spectral finiteness; neither supporting-line replacement nor analytic eigenvalue parameterization originates in this paper.

Robust policy counts. Finite-policy reductions based on convex representation appear in [10, Theorem 1] and [3]. Kurtz [13, Theorems 2 and 7] gives exactness and approximation bounds for objectives concave in the uncertain parameter over a convex uncertainty set, within a bounded integer-recourse setting. Here the candidate projectors form a continuum and $R(P, p) = \alpha(P) + pc(P) - F(p)$ is convex in price, so the required concavity is absent in general.

Lifting to the cost vector $(1, p, -F(p))$ does not repair that mismatch: its image is generally nonconvex, and convexifying it can change the menu objective. For example, the two coordinate presets of Model M have zero regret at every actual price. The lifted endpoint vectors are $(1, 0, 1)$ and $(1, 1, 0)$. At their midpoint $(1, 1/2, 1/2)$, both preset costs equal $1/2$. Thus the supremum over the lifted convex hull is at least $1/2$, although the original menu regret is zero. These policy-count results therefore do not directly supply (12) through a minimax swap. Conversely, our repair bound provides none of their algorithmic complexity guarantees. Continuous hitting and covering, including the farthest-reach characterization, already have explicit finite-adaptability treatments [5].

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